

CACEIS

Human Capital Valuation Methodology

Data Collection & Structuring — Parts 1 & 2 · Hackathon 2025

Part 1 — Ideal Data Ecosystem

Sources · Architecture · Framework · KPIs ·
Governance

Part 2 — Data Science Methodology

Good practices · Adaptive workflow · Starting
with minimal data

Part 1 — Ideal Data Ecosystem

Optimal human capital valuation system — designed from scratch

- Q1 — What data sources would you ideally collect (internal and external)?
- Q2 — What complementary data beyond the current scope would significantly strengthen your valuation methodology?

Internal & complementary data sources

Internal source	Data collected	Formula variable
DAP (WalkMe Pendo)	Click sequences, time/step, error rates, non-optimal paths — BU-level aggregation only, never individual	$Y_t \cdot \rho$
HRIS (SAP SF Workday)	Career trajectories, mobility, seniority, appraisals, absenteeism	P_t
LMS (Cornerstone, 360L)	Completion rates, scores, certifications, abandoned courses	$\lambda \cdot P_t$
Business KPIs (BI)	Error rates, SLA compliance, processing time, internal incidents	$Y_t \cdot \rho$
Workflow logs CISO	Validations, process bypasses, shadow AI detection patterns	ρ

External source	Business value	Comparability rule
Lightcast LinkedIn EG	Calibrates r — real market replacement cost, not an internal estimate	Measures posting demand, not actual supply. Use as 'tension index', cross-validate with CACEIS internal recruitment lead times.
WTW / Mercer / Aon	Contextualises BU scores — a λ score means nothing without a sector benchmark	Align turnover definition (voluntary vs. all), geographic perimeter, and reference year before any comparison.
AMF / CSSF / ESMA	Proactively triggers practice review flags before a regulatory incident occurs	Applicability varies by legal entity. Compliance filter required before auto-flagging practices 'under review'.
Institutional client NPS	External validation of Y_t — the most compelling ExCo evidence	Correlation $\neq$ causation. Control for incidents, market context, commercial relationship before attribution.
WEF / MGI / Gartner	Calibrates horizon T — skill half-life by function	Global macro-averages. Upper bound only; refine by asset servicing sub-function using internal depreciation data.

▸ Q3 — What collection architecture, tools, and governance would you implement?

Collection architecture & tools

Layer		Stack	Continuous data mechanism
C1 Capture	—	WalkMe/Pendo DAP · HRIS APIs · BI connectors	Real-time stream. Rolling windows: 30d operational / 90d learning agility / 12m retention.
C2 Platform	—	Airflow/dbt · pgvector · Mistral AI (FR sovereign cloud, GDPR-native)	Unanswered queries create a documentation backlog — the system learns from its own gaps.
C3 Valuation	—	Python scoring engine · D3.js visualisation	Regulatory publications auto-trigger IAS 36-style practice review flags.
C4 Activation	—	Optional sidebar UI · Mistral API · MS365 API (calendar + email)	Each improved behaviour feeds C1 — closing the feedback loop.

▸ Q4 — What is your conceptual framework for valuing human capital? How do you define and measure "value" beyond traditional cost accounting?

Conceptual framework — beyond cost accounting

Traditional accounting treats human capital as an expense — salaries appear in P&L, never on the balance sheet. The IFRS shift of the 2000s (IAS 36, IAS 38) established that firm value equals discounted future cash flows, not historical asset costs. Intangible assets now represent >80% of S&P 500 market value (Ocean Tomo, 2020). We apply the same logic to human capital: V_{HC} is the discounted value of future productive capacity, corrected for risk and amplified by learning.

$$V_{HC} = \Sigma [P_t \cdot Y_t^{base} \cdot \alpha_t \cdot \beta_t / (1+r)^t] \times (1-\rho) \times \lambda$$

Variable	Definition	Source in the system
$Y_t \cdot \alpha_t \cdot \beta_t$	Productivity = base output × personal AI amplifier (α) × collective platform multiplier (β). β grows at decreasing marginal cost — the more practices are documented, the more value each user extracts without additional AI spend. A collaborator in the system is worth $H_0 \cdot \alpha \cdot \beta$ vs. $H_0 \cdot \alpha$ without it.	C1 DAP · tool usage logs · reference adoption rate
P_t	Retention probability — function of engagement and training investment. Links HR care directly to asset preservation.	HRIS turnover history · engagement surveys
r	Discount rate = real market replacement cost (recruitment + onboarding + ramp-up). Grounded in Lightcast data, not internal estimates.	Lightcast + internal HR cost data
$\rho \in [0,1]$	Operational human risk — reduced by best-practice adoption and error rate decrease.	C3 BU scorecard
$\lambda > 1$ (capped)	Learning agility multiplier. Capped to prevent speculative inflation — the anti-IFRS-bubble guardrail.	C3 — novelty absorption speed per BU

IFRS parallel & guardrails

Like DCF, our model uses future flows. Four anti-speculation guardrails: (1) all variables are grounded in measured operational data, not projections; (2) BU-level aggregation prevents individual manipulation; (3) annual independent recalibration (auditor-equivalent); (4) automatic IAS 36-style depreciation when practices become obsolete.

► Q5 — What KPIs or analytical capabilities would this ideal ecosystem enable?

KPIs & analytical capabilities enabled

KPI	Decision enabled
Productivity gain from certified-sequence adoption (€/yr/BU) — <i>'certified' = expert-validated sequence minimising time AND error simultaneously, not 'optimal' abstractly</i>	Prioritise training investment by measured ROI, not intuition
Learning agility score — novelty absorption speed per BU (days to adoption)	Identify first-wave BUs for regulatory change rollouts
Knowledge concentration index — % critical practices mastered by <20% of staff (bus factor isolated from p)	Trigger targeted documentation before a key departure creates an irreversible crisis
β_t platform multiplier — value produced / marginal system cost	Demonstrate ROI of the collective platform vs. uncoordinated individual AI spend
Practice depreciation rate — % certified practices currently 'under review'	Detect BUs where human capital value erodes silently between audits
Exception corpus rate — % non-standardisable tasks per BU	Map precisely what is automatable vs. what requires irreplaceable human judgement — feeds strategic workforce planning

- Q6 — How would you ensure continuous data collection vs. one-time snapshots?
- Q7 — What technological infrastructure would support this ecosystem?
- Q8 — How would you ensure GDPR compliance, data quality, and ethical use?

Continuous collection · Infrastructure · Governance

Dimension	Response
Continuous vs. snapshot	DAP produces a permanent stream, never a monthly extraction. Three update regimes: <i>under observation</i> (continuous monitoring), <i>certified stable</i> (signal-triggered only — optimal practices don't need constant revision), <i>under review</i> (priority revalidation on regulatory or tool change). Unanswered Bibliothécaire queries auto-populate a documentation backlog.
Infrastructure	Mistral AI on French sovereign cloud (GDPR-native). pgvector on PostgreSQL for the vector store. Airflow + dbt for the data pipeline. Existing BI connectors. Optional sidebar UI overlaid on current tools — no new interface forced on users. MS365 API for training meeting auto-scheduling.
GDPR & ethics	Collective agreement with staff representatives (art. 6 RGPD) — more robust than legitimate interest alone. BU-level aggregation ≥5 people by default. Individual cards only under documented consent + art. 22 (purely advisory; all HR decisions remain human). Semi-annual bias audit by gender, age, seniority, geography. Annual independent model recalibration.
Data quality	Universal EDA audit on every new data source: coverage, missing values, definition alignment . External benchmarks explicitly labelled 'directional signal' or 'validated benchmark' depending on comparability checks passed. Explainability by design: "Why did my score change?" → sourced answer from the Bibliothécaire.

The 4-layer system — operational workflow

Each layer details its data sources, modules, and tools. Governance is transversal — it constrains all four layers simultaneously. The feedback loop connects activation back to capture.

COUCHE 1 — CAPTURE DES DONNÉES

Traçage des comportements et métadonnées au niveau agrégé B.U (jamais individuel)

Métadonnées usage

Séquences clavier ·
Chemins de navigation ·
Temps par tâche ·
Fréquence d'erreur

Données RH

SIRH · LMS ·
Formations suivies ·
Trajectoires de
carrière

KPIs business

Taux d'erreur ·
Respect SLA · NPS
client · Délai
traitement

Signaux faibles

Shadow AI ·
Contournements
processus · Pratiques
non documentées

↓ anonymisation · agrégation RGPD · pseudonymisation

COUCHE 2 — DATA PLATFORM (LA BIBLIOTHÈQUE)

Centralisation, normalisation, comparaison des pratiques — Mistral AI comme Bibliothécaire

Référentiel de pratiques

Bonnes pratiques certifiées ·
Pratiques à risque · Variantes
par tâche · Versioning continu

Moteur de comparaison

Même tâche · Finalités
identiques · Impacts distincts
sur KPIs

Le Bibliothécaire (Mistral)

Interface interrogeable · Accès
différencié collab/manager · RAG
sur pratiques

↓ scoring LLM · pondération criticité · modèle de valorisation

COUCHE 3 — VALORISATION DU CAPITAL HUMAIN

Cartes FIFA par B.U et par collaborateur · Modèle V_HC · Reporting financier

Scorecard B.U

Criticité compétence ·
Rareté · Risque
opérationnel ·
Learning agility ·
Partage savoir

Cartes FIFA B.U

Profil agrégé ·
Évolution temporelle ·
Benchmark inter-B.U
(Comex uniquement)

Carte individuelle

Miroir personnel ·
Consentement RGPD ·
Jamais notation
externe

Rapport V_HC (€)

Productivité gagnée ·
Risque réduit ·
Turnover évité · ROI
formation

↓ recommandations contextuelles · plans formation adaptés

COUCHE 4 — ACTIVATION TERRAIN

Assistant IA · Formations adaptatives · Manager insights — l'humain reste décideur

Fenêtre latérale (on-demand)

Ouverture volontaire ·
Reconnaissance du contexte
écran · Suggestion, jamais
surveillance

Formations adaptatives

Micro-learning · Généré par
Mistral · Capsules vidéo par
managers · Prestataires
orchestrés

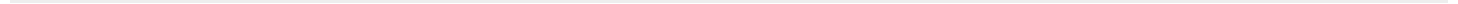
Manager insights

Alertes équipe · Plan d'action
ciblé · Suivi carte FIFA · MS365
auto-scheduling

↑ boucle de feedback — chaque action mesurée nourrit le modèle de valorisation

GOVERNANCE TRANSVERSALE (TOUTES COUCHES)

RGPD Accord collectif IRP · Anonymisation · Droit d'accès	Comité biais & équité Audit semestriel · Détection biais · Impact social	Contrôle humain Décisions RH humaines · IA consultative uniquement	Transparence Critères expliqués · Scores contestables · Recalibration annuelle
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Part 2 — Data Science Methodology

Working rigorously with unknown data — and starting with minimal information

The right stance: working with unknown data

Receiving an unknown dataset is the normal starting condition of any data science project. The challenge is to design a methodology that produces value regardless of what arrives, scales as more data becomes available, and always states clearly what the data does — and does not — allow us to conclude.

Sealed box principle

EDA is like receiving a sealed box: before assembling anything, you open it, inspect each piece, and understand what can actually be made with the available components. The goal is not to find what you expected — it is to find what the data genuinely reveals. Some hypotheses will be confirmed, some refuted, and some questions only emerge once the data is open. (EDA literature — *R for Data Science*, Hadley Wickham / Georgia Tech OMSCS 2026)

Starting with minimal data

With three columns — Employee ID, BU, and Seniority — you can already produce a meaningful analysis: knowledge concentration risk per BU, seniority distribution anomalies, and a first p estimate. You do not need a perfect dataset to begin. You need to know exactly what your available data allows you to conclude — and what it does not.

The 5-step process — applied to any HR dataset

1	2	3	4	5
Define hypotheses	Audit the data	EDA	Test hypotheses	Iterate
3 pre-defined hypotheses from V_HC theory, not from curiosity about the data	Coverage · quality · definitions · comparability — always, regardless of richness	Distributions · outliers · unexpected patterns — let the data shape the questions	Confirm or refute. Document what the data cannot answer — as important as what it can.	Enrich with new variables as they arrive. Each tier activates more formula variables.
Hypothesis-first	EDA-adaptive	Rigour	Honest limits	
Arrive with theory-grounded questions, not blank curiosity. Prevents confirmation bias.	Let the data shape which analyses are possible — never force an analysis the data can't support.	Compare only what is calculated the same way, on the same population, in the same period.	State clearly what the data does NOT allow to conclude — as rigorous as the findings themselves.	

Step 1 — Three pre-defined hypotheses from V_HC theory

Hypothesis	Expected finding	What it proves for the system
H1 — Training × performance	BUs with above-average training hours show measurably better performance or lower error proxies	Validates Y_t as a function of training investment — the core productivity link in the formula
H2 — Seniority × knowledge concentration	BUs with high seniority AND low training have critical practices concentrated in <20% of staff	Validates ρ as a calculable, BU-specific variable — not a generic estimate
H3 — Turnover × seniority band	Turnover concentrates in the 7–12 year seniority band — the knowledge transmission layer	Validates P_t as a predictive variable for knowledge loss, not just a lagging retention metric

Step 2 — Universal data audit (always, before any analysis)

Check	What we verify	Why it cannot be skipped
Coverage	Are all BUs represented? All geographies? What time window?	A dataset covering only one country invalidates cross-BU comparisons.
Completeness	% missing values per field — which fields are actually usable?	A field with 40% missing cannot anchor a hypothesis. Better to know before building on it.
Distributions	Histograms of seniority, training hours, performance scores. Outliers?	Skewed distributions affect correlation coefficients and can invert findings if undetected.
Definition alignment	How is turnover calculated? What does a score of 3 mean? What counts as a training hour?	compare only what is measured the same way. Document the definition before using the figure.

Steps 3–5 — The adaptive workflow by data tier

The analysis tier is determined by what the data actually contains — not by what we hoped to receive. Each tier is independently valuable. Tier 1 alone is sufficient to identify the pilot BU and present a preliminary business case.

Tier 1 — Minimum viable

Fields: Employee ID · BU / department · Seniority

Analyses: Knowledge concentration index · Seniority distribution shape · Bus factor risk per BU

Output: ρ proxy per BU · BU risk ranking · Pilot BU recommendation

Formula variables activated: First ρ estimate · Basis for DAP deployment case

Tier 2 — Standard HR dataset

+ Training hours · Completion rate · Turnover flag

Analyses: Training × seniority quadrant · Turnover by seniority band · Learning agility proxy

Output: λ and P_t per BU · Directional V_{HC} estimate · Training ROI hypothesis

Formula variables activated: λ , P_t calculated · First V_{HC} directional value

Tier 3 — Enriched dataset

+ Performance score · Mobility history · Engagement survey

Analyses: Training → performance correlation (BU-level) · Mobility × performance matrix · Full BU FIFA card draft

Output: Y_t proxy · Complete V_{HC} calculation · ExCo-ready business case with € estimates

Formula variables activated: Y_t proxy · All V_{HC} parameters calculable

Each tier is independently valuable — you do not need Tier 3 to begin. The system architecture does not change as data is enriched — it simply activates more formula variables.

What each tier activates in the 4-layer system

Tier	Layers activated	Business case produced
Tier 1	C1 partial · C3 (ρ proxy)	BU risk ranking. Pilot BU recommendation. Argument for DAP deployment to reach Tier 2.
Tier 2	C1 full · C3 ($\lambda + P_t$) · C4 training priorities	Directional V_HC per BU. Training ROI hypothesis. Learning agility gap between BUs. Strongest argument for C2 platform investment.
Tier 3	All 4 layers · Full V_HC formula	Training → performance correlation validated at BU level. Complete FIFA card draft. ExCo-ready business case with € estimates. POC complete.

The honest limits — what HR data proves vs. what it does not

What HR data proves	What it does NOT prove — and what is needed
Measurable gaps exist between BUs on training, seniority, turnover and knowledge concentration	That these gaps cause operational performance differences → business KPIs (error rates, SLA) required for causation
The V_HC formula parameters are calculable from available data	The exact monetary V_HC value → Y_t requires operational data to become a real productivity measure, not a proxy
A pilot BU can be identified with high confidence from risk/potential criteria	That the system will produce the projected ROI → demonstrated after C4 activation, not in the data analysis phase
BU-level aggregation is analytically more robust than individual-level	That individual FIFA cards are acceptable → requires a separate GDPR/IRP process

The real-world answer

Arrive with three pre-defined hypotheses from the theory. Run a universal EDA first. Apply the analysis tier the data actually supports. State clearly the boundary between what HR data proves and what business data would add. This is rigour under uncertainty: it does not require perfect data — it requires knowing exactly what your data does and does not allow you to conclude.

Part 2 will be completed with real findings once the anonymised CACEIS dataset is received. The methodology above applies unchanged — tier determination, audit, and hypothesis testing run as described. The workflow and formula remain the analytical backbone regardless of data richness.